Reliability Analysis of Mechanical Springs

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Abstract

This study evaluates the mechanical durability of springs, focusing on the quantification of "*Cycles to Failure*" as a reliability metric. Leveraging a Full Factorial experimental design, it investigates the impact of three factors: *mechanical stroke length*, *processing temperature*, and *manufacturing method*. Because the experimental response was right-censored at 5,000 cycles, survival analysis techniques – specifically Kaplan-Meier survival curves and Cox Proportional Hazards (CPH) modeling – were employed to account for units that survived the testing duration.

The analysis identifies mechanical stress (stroke) and manufacturing technique as the main drivers of component longevity. The optimized Cox model reveals that the "New" manufacturing method provides a statistically significant 89% reduction in the hazard of failure compared to the "Old" method. Furthermore, each additional mil of stroke length increases the risk of failure by approximately 17%. While processing temperature was statistically significant, its impact was found to be less important from a practical standpoint (Hazard Ratio HR ≈ 1.002).

Model diagnostics, including Martingale residuals, DFBETAS and Schoenfeld tests (Global p = 0.82), validated the model. These findings provide an argument to adopt the "New" manufacturing process and optimize design parameters for lower stroke lengths to maximize reliability.

R code and dataset

This study was performed using R version 4.5.2 and is available as an R-Markdown file at SamdGuizani/DSTI_Survival_Analysis. A copy of the studied dataset is also available in the same repository. It is a publicly available dataset (refer to section 1.3 for details).

1 Introduction

1.1 Context

In mechanical engineering, the ability to accurately predict "Cycles to Failure" of components is critical information. High-reliability components are essential for reducing warranty liabilities, minimizing industrial downtime, and ensuring safety. By quantifying the degradation over time

of components like mechanical springs, organizations can transition from reactive to proactive maintenance. Moreover, this knowledge allows engineers to identify stress factors early in the development cycle, ultimately designing better longevity products.

1.2 Problem Statement

The objective of this analysis is to isolate and quantify the influence of mechanical stress, environmental conditions, and manufacturing parameters on spring durability. Specifically, the study seeks to determine how varying mechanical displacement (*A.Stroke*), processing temperature (*B.ProcessTpt*), and the manufacturing technique (*C.Method*) impact the survival probability of the components. The goal is to produce a validated risk statistical model.

1.3 Data Provenance

The data utilized in this study is sourced from Meeker et al. [1] , hosted within the *Statistical Methods for Reliability Data* repository at Iowa State University [2] . This dataset represents a controlled factorial experiment to evaluate failure-time data under varying conditions.

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2 Methods

2.1 Experimental Design & Variables

The study uses a Full Factorial Design with 3 factors, and 9 replicates for every factor combination, resulting in a total number of observations of N=108. Table 1 introduces the response and the factors of the experiment.

Table 1 – Covariates and response in the spring reliability study

Variables	Type	Description
A.Stroke	Covariate, Numeric	Mechanical displacement tested at 3 levels: 50, 60, and 70 mils.
B.ProcessTpt	Covariate, Numeric	Processing temperature tested at 2 levels: 500 and 1000 °F
C.Method	Covariate, Categorical	Manufacturing technique tested at 2 levels: "Old" vs. "New".
Cycles to Failure	Response, Numeric	Time-to-event variable, right-censored at 5,000 cycles.

2.2 Rationale for Survival Analysis

Standard linear regression models are insufficient for reliability data because they cannot accommodate "censored" observations. In this study, springs surviving beyond 5,000 cycles are considered right-censored. Survival analysis, on the other hand, provides the adequate framework to study such data.

2.3 Statistical Study Workflow

The analysis followed a multi-stage workflow to ensure both baseline understanding and predictive accuracy:

1. **Exploratory Data Analysis (EDA):** Table summaries and graphical data exploration were used to obtain a general understanding of the data and formulate hypotheses about the influencing factors on the spring life duration.
2. **Non-Parametric Analysis:** Kaplan-Meier (KM) estimation was used to visualize survival curves. Stratified Log-Rank tests were conducted to identify significant factors while controlling confounding variables.
3. **Semi-Parametric Modeling:** A Cox Proportional Hazards (CPH) regression was fitted to quantify the Hazard Ratios (HR) for each factor. An initial model included the covariates and all their two-way interactions (Stroke:ProcessTpt, Stroke:Method, ProcessTpt:Method). Automated stepwise selection using the Akaike Information Criterion (AIC) was later used to build a more parsimonious model.
4. **Diagnostics and Model Assumptions Assessment:** The selected CPH model was verified through three diagnostic tools:
   - **Martingale residuals:** To validate the functional form of the model (i.e., assess if mathematically transforming the covariates could provide better predictive models).
   - **DFBETAS residuals:** To check for highly influential observations, skewing hazard estimates.
   - **Schoenfeld residuals and log-log survival plots:** To check the *Proportional Hazards Assumption* (CPH models are built on the assumption that the impact of covariates remains constant over time).

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3 Results

3.1 Exploratory Data Analysis (EDA)

A noticeable fact in the data is the proportion of censored observations: 35 observations among 108 (~32%) are censored. The effect of censoring on the Cycles response distribution is observable on the histogram in Figure 1, showing a right-bounded distribution with high count of observations at 5000 cycles.

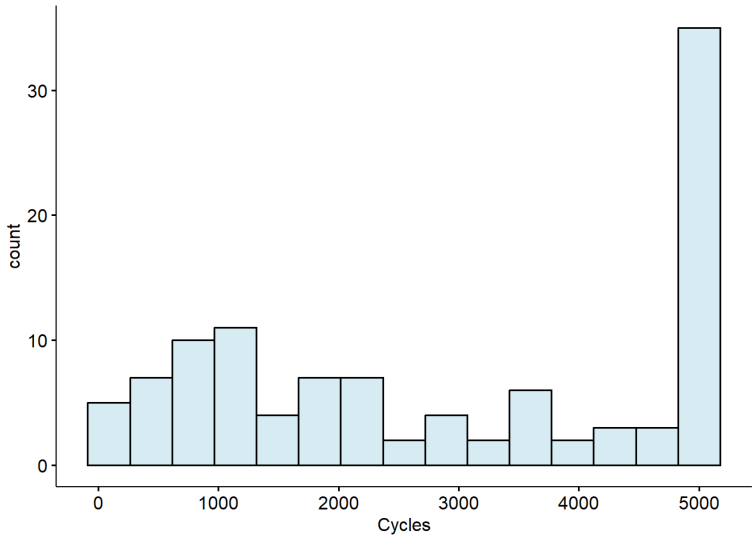

Figure 1 – Histogram of Cycles response distribution

Boxplots (Figure 2) indicated that A.Stroke and C.Method may be the main drivers of failure. Notably, the "Old" manufacturing method showed a median failure time of approximately 1,500 cycles, whereas the "New" method's median approached the 5,000-cycle censoring threshold. B.ProcessTpt shows a less pronounced visual impact, suggesting its influence might be secondary to mechanical stress.

Characteristic	N = 108[1]
Cycles	2,814 (1,189, 5,000)
A.Stroke	
50	36 (33%)
60	36 (33%)
70	36 (33%)
B.ProcessTpt	
500	54 (50%)
1000	54 (50%)
C.Method	
Old	54 (50%)
New	54 (50%)
Censoring	73 (68%)

[1] Median (Q1, Q3); n (%)

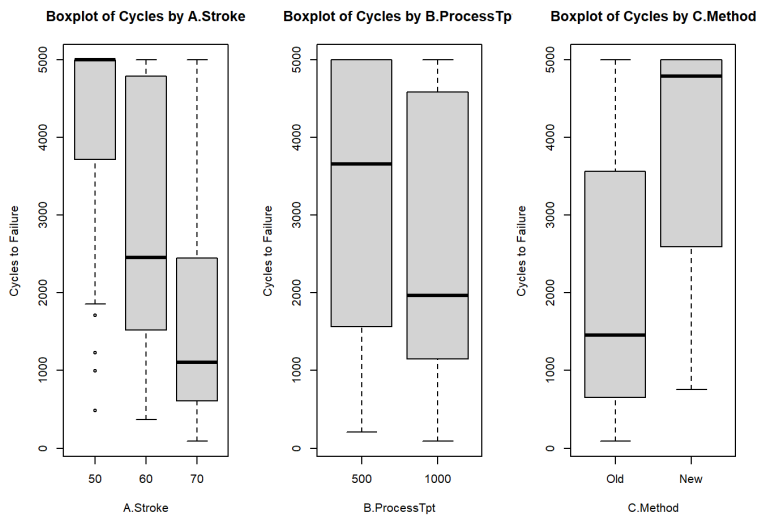

Figure 2 – Summary of variables and Cycles boxplots vs. covariates

3.2 Non-Parametric Analysis

Kaplan-Meier survival curves (Figure 3) show that covariates A.Stroke and C.Method are statistically significant ($p < 0.001$), while B.ProcessTpt is marginally failing the 0.05 significance threshold.

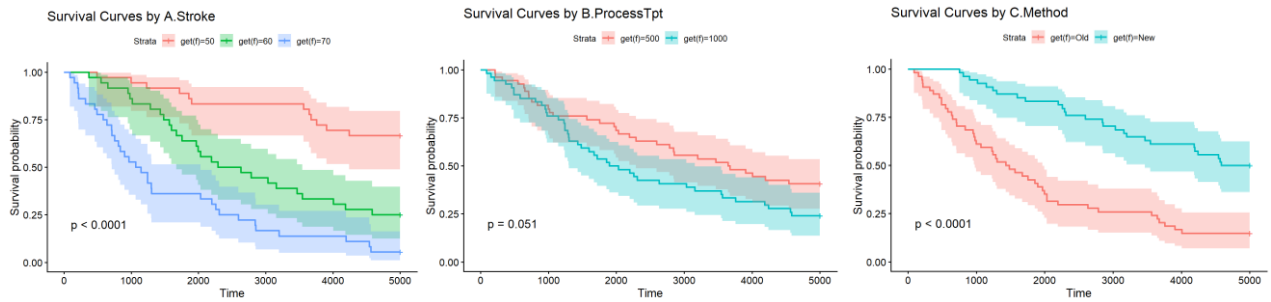

Figure 3 – Kaplan Meier survival curves for Cycles vs. covariates

Log-Rank tests were conducted on each covariate (Table 2). For each test, the other covariates than the tested one were used as stratification variables. Besides confirming the statistical significance of A.Stroke and C.Method effects on the response ($p < 0.001$), the stratified Log-Rank on B.ProcessTpt was found to be statistically significant as well ($p = 0.002$).

Table 2 – Log-rank test results

Log-rank test for covariates	p-value
A.Stroke `Formula: Surv(Cycles, Censoring) ~ A.Stroke + strata(B.ProcessTpt, C.Method)`	< 0.001
B.ProcessTpt `Formula: Surv(Cycles, Censoring) ~ B.ProcessTpt + strata(A.Stroke, C.Method)`	0.002
C.Method `Formula: Surv(Cycles, Censoring) ~ C.Method + strata(A.Stroke, B.ProcessTpt)`	< 0.001

3.3 Semi-parametric Cox Proportional Hazards (CPH) Model

A first CPH model, including all covariates and their interactions has been fitted:

```
Surv(Cycles, Censoring) ~ A.Stroke + B.ProcessTpt + C.Method + A.Stroke * B.ProcessTpt +
A.Stroke * C.Method + B.ProcessTpt * C.Method
```

All terms were found to be statistically insignificant at significance level 0.05, except A.Stroke ($p < 0.001$), hence calling for model term selection.

The final model was obtained through stepwise AIC selection and contains only the terms for main effects:

```
Surv(Cycles, Censoring) ~ A.Stroke + B.ProcessTpt + C.Method.
```

Table 3 reports the Hazard Ratios (HR) for each covariate included in the model. The main driver to improve spring reliability was found to be using the “New” C.Method. Furthermore, reducing the mechanical stress (A.Stroke) reduces the risk of units failing. B.ProcessTpt was found to marginally impact the spring’s life duration.

Table 3 – Final CPH model Hazard Ratio vs. covariates

Covariate	Hazard Ratio (HR)	Risk Quantification
A.Stroke	1.169	~17% risk increase per 1 mil increase in stroke length.
C.Method (New)	0.110	~89% risk reduction compared to "Old" method.
B.ProcessTpt	1.002	Significant ($p < 0.01$) but negligible effect.

3.4 Diagnostics and CPH Model Assumptions Assessment

3.4.1 Martingale residuals assessment

Plotting the final model Martingale residuals vs. covariates (Figure 4) shows a comparable spread at the various tested levels. However, checking the shape of the relationship between the residuals and the continuous numeric covariates is not easy in this case, given these covariates were only set at discrete levels, within the context of a Full Factorial design.

From this graphical assessment, showing an absence of clear pattern, transformations on the covariates are deemed unnecessary and the linear specification for numeric covariates in the model is an adequate modelling choice.

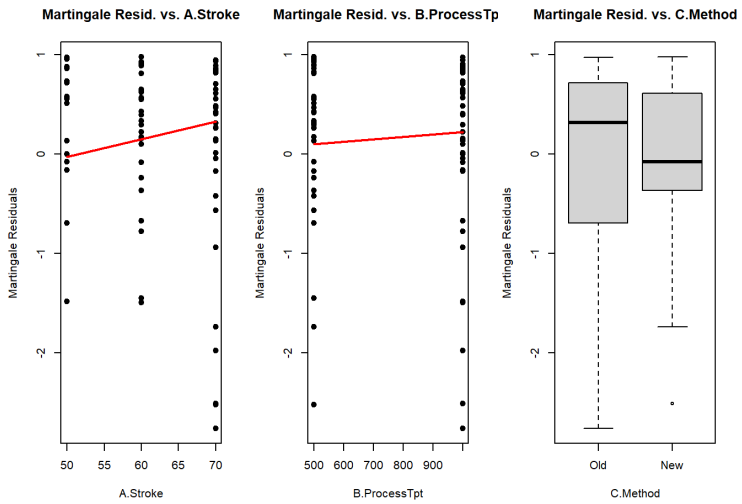

Figure 4 – Martingale residuals vs. covariates

3.4.2 DFBETAS residuals assessment

Plotting the DFBETAS residuals vs observation index (Figure 5) does not clearly point out highly influential observations.

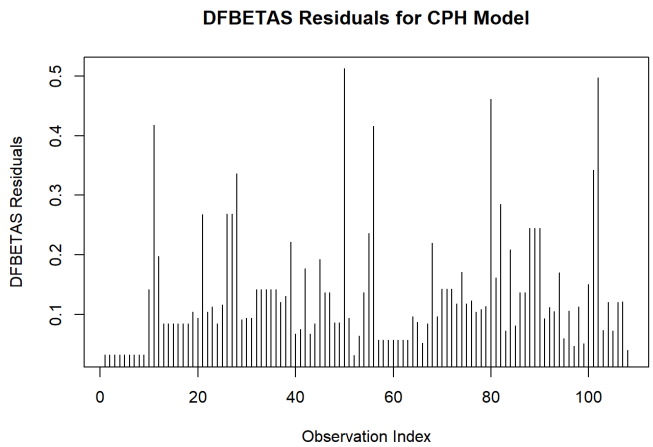

Figure 5 – DFBETAS residuals vs. observation index

3.4.3 Proportional Hazard (PH) Assumption Verification

3.4.3.1 Shoenfeld residuals assessment

The Shoenfeld residual test was conducted on each covariate and for the global model. The results (Table 4) show non-significant p-values, supporting the assumption of proportional hazards.

Table 4 – Shoenfeld residuals tests for each covariate and for final CPH model

	X^2	df	p
A.Stroke	0.0923	1	0.76
B.ProcessTpt	0.5749	1	0.45
C.Method	0.3439	1	0.56
GLOBAL	0.9118	3	0.82

Additionally, Shoenfeld residuals have been plotted against time for each covariate (Figure 6). The flat behavior of Shoenfeld residuals across time is a further indication of hazard proportionality.

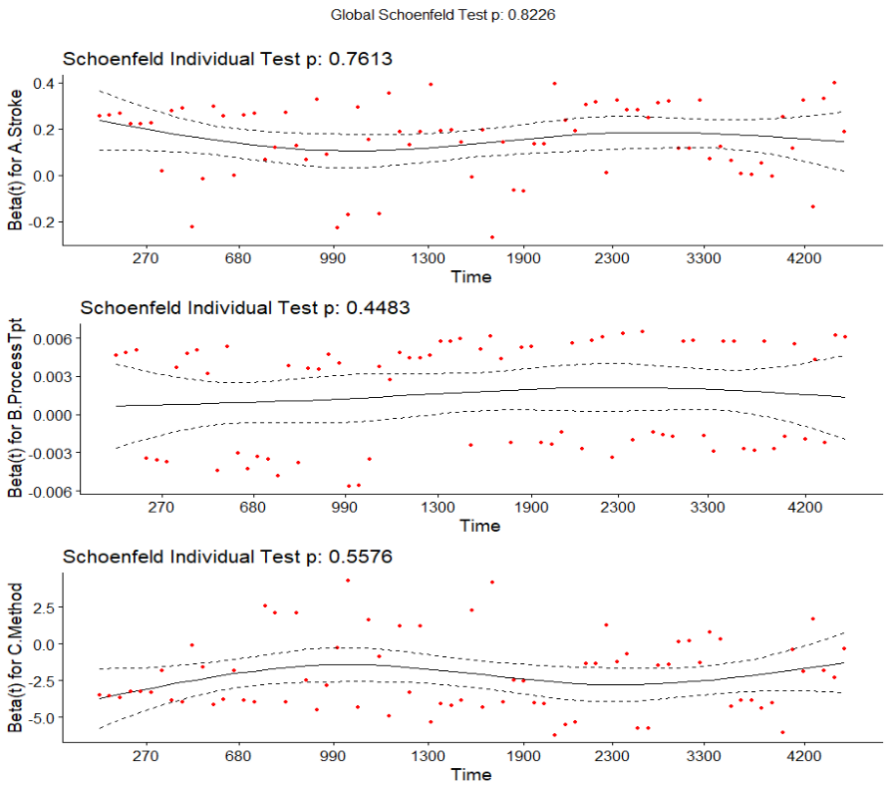

Figure 6 – Shoenfeld residuals vs. Time for each covariate

3.4.3.2 Log-log survival curves

Another approach to assess proportional hazards is to compare the model predicted survival curves at each level of a given covariate, using log-log plots. This approach is valid for categorical covariate. However, in this study, as the numeric covariates are varied at discrete levels, the curves were produced for the 3 variables (Figure 7). For each subplot, the survival curves are created by changing a chosen covariate while keeping all the others at fixed levels.

The log-log survival curve plots show approximately parallel lines which support the proportional hazard assumption.

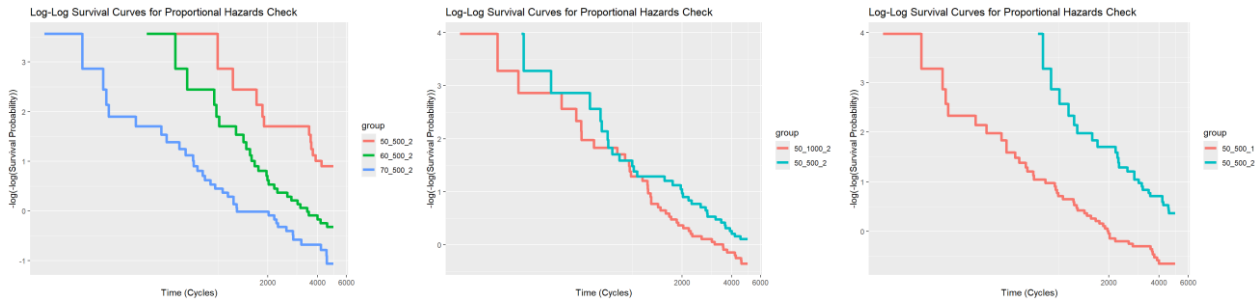

Figure 7 – Log-log survival curves (from left to right: A.Stroke, B.ProcessTpt and C.Method)

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4 Discussion

4.1 Major Conclusions

Integrating survival analysis into mechanical testing provides a mathematically sound direction for enhancing product durability. This study gives engineering teams insights to make design decisions rooted in statistical quantification rather than empirical guesswork. The following recommendations can be derived from the analysis:

1. **The "New" Manufacturing Method Advantage:** The "New" method is the most effective lever for increasing reliability, providing 89% reduction in failure risk. Adoption of this process should be prioritized for quality assurance.

2. **Stress Sensitivity:** Stroke length is the predominant failure-driving stressing factor. The 17% failure risk increase per mil indicates that springs are significantly sensitive to displacement. Therefore, focusing the design on reducing mechanical stroke is recommended, rather than acting on the processing temperature (temperature variations have minimal practical impact on component failure).

4.2 Methodological Comparison

While the non-parametric Log-Rank tests identify the impacting factors, the semi-parametric CPH model allows access to Hazard Ratios. These metrics are required for predictive maintenance, warranty modeling and for calculating the return on investment for new manufacturing technologies.

4.3 Limitations & Outlook

The current analysis focused on using non-parametric and semi-parametric modelling approaches. In particular, the semi-parametric Cox Proportional Hazard (CPH) was a suitable choice, which offers the benefit of not requiring to specify the hazard function shape.

Nevertheless, future research could investigate parametric models such as Accelerated Failure Time (AFT), leveraging prior knowledge to adequately specify the hazard function, and comparing their performance to the CPH model.

References

[1] William Q. Meeker, Luis A. Escobar, Francis G. Pascual (2021), Statistical Methods for Reliability Data, 2nd Edition (2021), John Wiley & Sons Inc.

[2] https://iastate.figshare.com/articles/dataset/Reliability_Data_Laboratory_Failure-Time_and_Single-Time_Strength_Data/14454753?file=28330338